

# ECE 332 — Homework 3: Generated Answers from Final Cheat Sheet

Each solution shows the cheat sheet snippet used, then the answer.

## Problem 1 — LHC Wave Normally Incident on a Dielectric

20 pts

**Setup:** 200 MHz LHC plane wave in air,  $|E| = 5$  V/m, normally incident on dielectric ( $\epsilon_r = 4$ ) at  $z \geq 0$ . Positive max at  $z = 0$ ,  $t = 0$ .

### NORMAL INCIDENCE — $\Gamma$ AND $\tau$

Reflection coeff. *boundary at  $z = 0$ , wave from medium 1*

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \text{ (unitless)}$$

Transmission coeff. *always*

$$\tau = 1 + \Gamma = \frac{2\eta_2}{\eta_2 + \eta_1}$$

Lossless  $\eta_i$   $\sigma = 0$

$$\eta_i = \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)} \quad \eta_0 = 120\pi \approx 377 \Omega$$

Low-loss  $\eta_2$  (poor conductor)  $\sigma/(\omega\epsilon) \ll 1$

$$\eta_2 \approx \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon}\right) \text{ (}\Omega\text{)}$$

*For boundary  $\Gamma/\tau$ , take just  $|\eta_2| \approx \eta_0/\sqrt{\epsilon_{r2}}$ ; the  $j$  term is tiny.*

### POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100 |\Gamma|^2$$

% transmitted  *$\eta$ -ratio matters!*

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check:  $\%P_r + \%P_t = 100$ .

*If medium 2 is denser ( $\eta_2 < \eta_1$ ),  $|\tau| > 1$  but  $\%P_t < 100\%$  because of the  $\eta_1/\eta_2$  factor.*

### $\tilde{\mathbf{E}}$ PHASOR TEMPLATES

Wave in  $+\hat{z}$ , incident has unit-vector pol.  $\hat{e}$ , amplitude  $a_0$ :

$$\tilde{\mathbf{E}}^i = a_0 \hat{e} e^{-jk_1 z}$$

$$\tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{+jk_1 z} \text{ (travels in } -\hat{z}\text{)}$$

$$\tilde{\mathbf{E}}^t = \tau a_0 \hat{e} e^{-jk_2 z} \text{ (lossless medium 2)}$$

$$\tilde{\mathbf{E}}^t = \tau a_0 \hat{e} e^{-\alpha_2 z} e^{-j\beta_2 z} \text{ (lossy medium 2)}$$

Total in medium 1:  $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$ .

$\hat{e}$  by polarization *plug into templates above*

| Pol.               | $\hat{e}$                               | Note              |
|--------------------|-----------------------------------------|-------------------|
| Lin. $\hat{x}$     | $\hat{x}$                               | $E_y = 0$         |
| Lin. $\hat{y}$     | $\hat{y}$                               | $E_x = 0$         |
| Lin. at $45^\circ$ | $(\hat{x} + \hat{y})/\sqrt{2}$          | in phase          |
| RHC                | $\hat{x} - j\hat{y}$                    | $\delta = -\pi/2$ |
| LHC                | $\hat{x} + j\hat{y}$                    | $\delta = +\pi/2$ |
| General            | $a_x \hat{x} + a_y e^{j\delta} \hat{y}$ | elliptical        |

$\Gamma, \tau$  act on  $\hat{e}$  unchanged. Only swap the polarization, not the formulas.

## Wave parameters in medium 1 (air):

Angular frequency from  $f$ :

$$\omega = 2\pi f = 2\pi(2 \times 10^8) = 4\pi \times 10^8 \text{ rad/s}$$

Wavenumber from the lossless plane wave formula  $k = \omega\sqrt{\epsilon_r}/c$ . In air,  $\epsilon_r = 1$  so this reduces to  $k_1 = \omega/c$ :

$$k_1 = \frac{\omega}{c} = \frac{4\pi \times 10^8}{3 \times 10^8} = \frac{4\pi}{3} \text{ rad/m}$$

**(a) LHC incident phasor.** The PHASOR TEMPLATES section says the incident wave has the form  $\tilde{\mathbf{E}}^i = a_0 \hat{e} e^{-jk_1 z}$  where  $\hat{e}$  depends on polarization. From the polarization table:

$$\text{LHC} \Rightarrow \hat{e} = \hat{x} + j\hat{y}$$

With  $a_0 = 5$  V/m (the modulus) and  $k_1 = 4\pi/3$ :

$$\tilde{\mathbf{E}}^i = 5(\hat{x} + j\hat{y}) e^{-j4\pi z/3} \text{ V/m}$$

**(b) Reflection coefficient  $\Gamma$  and transmission coefficient  $\tau$ .**

Original formula for intrinsic impedance:  $\eta = \sqrt{\mu/\varepsilon}$ .

Medium 1 (air, free space,  $\mu = \mu_0$  and  $\varepsilon = \varepsilon_0$ ):

$$\eta_1 = \eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = \sqrt{(4\pi \times 10^{-7})(36\pi \times 10^9)} = 120\pi \Omega \approx 377 \Omega$$

Medium 2 (nonmagnetic dielectric,  $\mu_2 = \mu_0$ ,  $\varepsilon_2 = \varepsilon_r \varepsilon_0$ ). The general  $\eta = \sqrt{\mu/\varepsilon}$  simplifies to the lossless shortcut  $\eta = \eta_0/\sqrt{\varepsilon_r}$ :

$$\eta_2 = \sqrt{\frac{\mu_0}{\varepsilon_r \varepsilon_0}} = \frac{1}{\sqrt{\varepsilon_r}} \sqrt{\frac{\mu_0}{\varepsilon_0}} = \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{120\pi}{\sqrt{4}} = \frac{120\pi}{2} = 60\pi \Omega$$

Now apply the boundary formula  $\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$ :

$$\Gamma = \frac{60\pi - 120\pi}{60\pi + 120\pi} = \frac{-60\pi}{180\pi} = \boxed{-\frac{1}{3}}$$

$$\tau = 1 + \Gamma = 1 - \frac{1}{3} = \boxed{\frac{2}{3}}$$

**(c) Reflected, transmitted, and total field phasors.**

Wavenumber in medium 2, again from  $k = \omega\sqrt{\varepsilon_r}/c$  but now  $\varepsilon_r = 4$ :

$$k_2 = \frac{\omega\sqrt{4}}{c} = 2k_1 = \frac{8\pi}{3} \text{ rad/m}$$

From the PHASOR TEMPLATES (reflected wave travels in  $-\hat{z}$ , sign of  $z$ -term flips):

$$\tilde{\mathbf{E}}^r = \Gamma \cdot a_0 \hat{e}^{+jk_1 z} = -\frac{1}{3} \cdot 5(\hat{x} + j\hat{y}) e^{+j4\pi z/3} = -\frac{5}{3}(\hat{x} + j\hat{y}) e^{+j4\pi z/3}$$

Transmitted wave in lossless medium 2:

$$\tilde{\mathbf{E}}^t = \tau \cdot a_0 \hat{e}^{-jk_2 z} = \frac{2}{3} \cdot 5(\hat{x} + j\hat{y}) e^{-j8\pi z/3} = \frac{10}{3}(\hat{x} + j\hat{y}) e^{-j8\pi z/3}$$

Total field in medium 1 is incident + reflected:

$$\boxed{\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r = 5(\hat{x} + j\hat{y}) \left( e^{-j4\pi z/3} - \frac{1}{3} e^{+j4\pi z/3} \right) \text{ V/m}}$$

**(d) Percentages of incident average power reflected and transmitted.**

From the POWER REFLECTION snippet,  $\%P_r = 100|\Gamma|^2$  and  $\%P_t = 100|\tau|^2(\eta_1/\eta_2)$ . The  $\eta_1/\eta_2$  factor accounts for the change in wave impedance: even though  $|\tau| > 0$ , the medium beyond carries power differently.

$$\%P_r = 100 \cdot \left| -\frac{1}{3} \right|^2 = 100 \cdot \frac{1}{9} = \boxed{11.1\%}$$

$$\%P_t = 100 \cdot \left| \frac{2}{3} \right|^2 \cdot \frac{120\pi}{60\pi} = 100 \cdot \frac{4}{9} \cdot 2 = \boxed{88.9\%}$$

Sanity check:  $11.1\% + 88.9\% = 100\%$ . Energy conserved.

## Problem 2 — Same Wave, Poor-Conductor Medium 2

20 pts

**Setup:** Replace medium 2 with poor conductor:  $\epsilon_r = 2.25$ ,  $\mu_r = 1$ ,  $\sigma = 10^{-4}$  S/m. Everything else same as P1.

NORMAL INCIDENCE —  $\Gamma$  AND  $\tau$ 

Reflection coeff. *boundary at  $z = 0$ , wave from medium 1*

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \text{ (unitless)}$$

Transmission coeff. *always*

$$\tau = 1 + \Gamma = \frac{2\eta_2}{\eta_2 + \eta_1}$$

Lossless  $\eta_i$   $\sigma = 0$

$$\eta_i = \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \text{ (}\Omega\text{)} \quad \eta_0 = 120\pi \approx 377 \text{ }\Omega$$

Low-loss  $\eta_2$  (poor conductor)  $\sigma/(\omega\epsilon) \ll 1$

$$\eta_2 \approx \sqrt{\frac{\mu}{\epsilon}} \left( 1 + j \frac{\sigma}{2\omega\epsilon} \right) \text{ (}\Omega\text{)}$$

For boundary  $\Gamma/\tau$ , take just  $|\eta_2| \approx \eta_0/\sqrt{\epsilon_{r2}}$ ; the  $j$  term is tiny.

## LOW-LOSS QUICK PARAMS (med 2)

Check first  $\sigma/(\omega\epsilon)$

$$\alpha_2 \approx \frac{\sigma_2}{2} \sqrt{\frac{\mu_2}{\epsilon_2}} = \frac{\sigma_2 \eta_0}{2\sqrt{\epsilon_{r2}}} \text{ (Np/m)}$$

$$\beta_2 \approx \omega \sqrt{\mu_2 \epsilon_2} = \frac{\omega \sqrt{\epsilon_{r2}}}{c} \text{ (rad/m)}$$

$$\eta_2 \approx \frac{\eta_0}{\sqrt{\epsilon_{r2}}} \angle \approx 0$$

$$\epsilon = \epsilon_r \epsilon_0; \epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}; 36\pi\epsilon_0 \approx 1 \times 10^{-9}.$$

## POWER REFLECTION &amp; TRANSMISSION

% reflected *always*

$$\%P_r = 100 |\Gamma|^2$$

% transmitted  *$\eta$ -ratio matters!*

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check:  $\%P_r + \%P_t = 100$ .

If medium 2 is denser ( $\eta_2 < \eta_1$ ),  $|\tau| > 1$  but  $\%P_t < 100\%$  because of the  $\eta_1/\eta_2$  factor.

(a) Medium 1 (air) is unchanged from P1. Same incident phasor:

$$\eta_1 = 120\pi \text{ }\Omega, \quad k_1 = \frac{4\pi}{3} \text{ rad/m}$$

$$\tilde{\mathbf{E}}^i = 5(\hat{x} + j\hat{y}) e^{-j4\pi z/3} \text{ V/m}$$

(b) Classify medium 2 first. The LOSSY classification needs the loss tangent  $\sigma/(\omega\epsilon)$ :

$$\frac{\sigma_2}{\omega\epsilon_2} = \frac{\sigma_2}{\omega\epsilon_r\epsilon_0} = \frac{10^{-4}}{(4\pi \times 10^8)(2.25)(8.854 \times 10^{-12})} = 4 \times 10^{-3}$$

This is  $\ll 1$ , so medium 2 is a **low-loss dielectric**. Use the LOW-LOSS QUICK PARAMS approximations, not the full lossy formulas.

Attenuation constant (small but nonzero now):

$$\alpha_2 \approx \frac{\sigma_2 \eta_0}{2\sqrt{\epsilon_{r2}}} = \frac{10^{-4} \cdot 120\pi}{2\sqrt{2.25}} = \frac{0.0377}{3} = 1.26 \times 10^{-2} \text{ Np/m}$$

Phase constant (same as lossless approximation since  $\sigma$  is tiny):

$$\beta_2 \approx \frac{\omega \sqrt{\epsilon_{r2}}}{c} = \frac{(4\pi \times 10^8)(1.5)}{3 \times 10^8} = 2\pi \text{ rad/m}$$

Impedance: low-loss approximation gives  $\eta_c \approx \eta_0/\sqrt{\epsilon_r}$  (essentially real for this small loss tangent):

$$\eta_2 \approx \frac{\eta_0}{\sqrt{\epsilon_{r2}}} = \frac{120\pi}{1.5} = 80\pi \text{ }\Omega$$

Now apply the boundary formulas exactly as in P1:

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{80\pi - 120\pi}{80\pi + 120\pi} = \frac{-40\pi}{200\pi} = \boxed{-\frac{1}{5}}$$

$$\tau = 1 + \Gamma = \boxed{\frac{4}{5}}$$

(c) **Field phasors.** Same templates as P1 except the transmitted wave now has the lossy decay factor  $e^{-\alpha_2 z}$ :

$$\tilde{\mathbf{E}}^r = \Gamma \cdot 5(\hat{x} + j\hat{y}) e^{+j4\pi z/3} = -(\hat{x} + j\hat{y}) e^{+j4\pi z/3} \text{ V/m}$$

$$\tilde{\mathbf{E}}^t = \tau \cdot 5(\hat{x} + j\hat{y}) e^{-\alpha_2 z} e^{-j\beta_2 z} = 4(\hat{x} + j\hat{y}) e^{-1.26 \times 10^{-2} z} e^{-j2\pi z} \text{ V/m}$$

$$\boxed{\tilde{\mathbf{E}}_1 = 5(\hat{x} + j\hat{y}) \left( e^{-j4\pi z/3} - \frac{1}{5} e^{+j4\pi z/3} \right) \text{ V/m}}$$

(d) **Power percentages.** Same formulas as P1:

$$\%P_r = 100|\Gamma|^2 = 100 \cdot \frac{1}{25} = \boxed{4\%}$$

$$\%P_t = 100|\tau|^2 \frac{\eta_1}{\eta_2} = 100 \cdot \frac{16}{25} \cdot \frac{120\pi}{80\pi} = 100 \cdot \frac{16}{25} \cdot \frac{3}{2} = \boxed{96\%}$$

*Sanity check:*  $4\% + 96\% = 100\%$ . The dielectric is denser than air ( $\epsilon_r = 2.25$  vs 1), so most of the wave gets through.

### Problem 3 — Parallel-Polarized Wave Through Two Dielectric Layers 10 pts

**Setup:**  $\theta_i = 30^\circ$  from air, layer 1 ( $\epsilon_r = 6.25$ ,  $t = 5$  cm), layer 2 ( $\epsilon_r = 2.25$ ,  $t = 5$  cm), exits to air.

| SNELL'S LAW — OBLIQUE INCIDENCE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | LATERAL DISPLACEMENT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |          |     |        |     |        |     |     |        |        |      |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------|-----|--------|-----|--------|-----|-----|--------|--------|------|-----|
| <p><b>Single boundary</b> <i>angles from normal</i></p> $\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$ <p><b>Multilayer chain</b> <i>stack of slabs 1 <math>\rightarrow</math> 2 <math>\rightarrow</math> 3 <math>\rightarrow</math> 4</i></p> $\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$ <p><b>Air-slabs-air shortcut</b> <i>outer media identical</i></p> $\theta_{\text{exit}} = \theta_{\text{incident}}$ <p><i>Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).</i></p> | <p><b>Beam offset through <math>N</math> slabs</b> <i>thickness <math>t_i</math>, refracted angle <math>\theta_i</math></i></p> $d = \sum_{i=1}^N t_i \tan \theta_i$ <p><i>Each slab adds <math>t_i \tan \theta_i</math>. Final air-to-air exit returns to <math>\theta_1</math> but is laterally shifted by <math>d</math>.</i></p> <p><b>Common values</b> <i>arctan</i></p> <table border="1" style="margin-left: auto; margin-right: auto; border-collapse: collapse;"> <thead> <tr style="background-color: #e0f2f1;"> <th style="padding: 2px 5px;">sin <math>\theta</math></th> <th style="padding: 2px 5px;"><math>\theta</math></th> </tr> </thead> <tbody> <tr><td style="text-align: center;">0.2</td><td style="text-align: center;">11.54°</td></tr> <tr><td style="text-align: center;">1/3</td><td style="text-align: center;">19.47°</td></tr> <tr><td style="text-align: center;">1/2</td><td style="text-align: center;">30°</td></tr> <tr><td style="text-align: center;">0.5774</td><td style="text-align: center;">35.26°</td></tr> <tr><td style="text-align: center;">1/√2</td><td style="text-align: center;">45°</td></tr> </tbody> </table> | sin $\theta$ | $\theta$ | 0.2 | 11.54° | 1/3 | 19.47° | 1/2 | 30° | 0.5774 | 35.26° | 1/√2 | 45° |
| sin $\theta$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\theta$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |              |          |     |        |     |        |     |     |        |        |      |     |
| 0.2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 11.54°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |          |     |        |     |        |     |     |        |        |      |     |
| 1/3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 19.47°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |          |     |        |     |        |     |     |        |        |      |     |
| 1/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 30°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |              |          |     |        |     |        |     |     |        |        |      |     |
| 0.5774                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 35.26°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |              |          |     |        |     |        |     |     |        |        |      |     |
| 1/√2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 45°                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |              |          |     |        |     |        |     |     |        |        |      |     |

(a) Chained Snell's law:

$$\sin \theta_2 = \sin 30^\circ \sqrt{\frac{1}{6.25}} = 0.5 \cdot 0.4 = 0.2 \Rightarrow \theta_2 = \sin^{-1}(0.2) = \boxed{11.54^\circ}$$

$$\sin \theta_3 = 0.2 \sqrt{\frac{6.25}{2.25}} = 0.2 \cdot \frac{5}{3} = \frac{1}{3} \Rightarrow \theta_3 = \sin^{-1}(1/3) = \boxed{19.47^\circ}$$

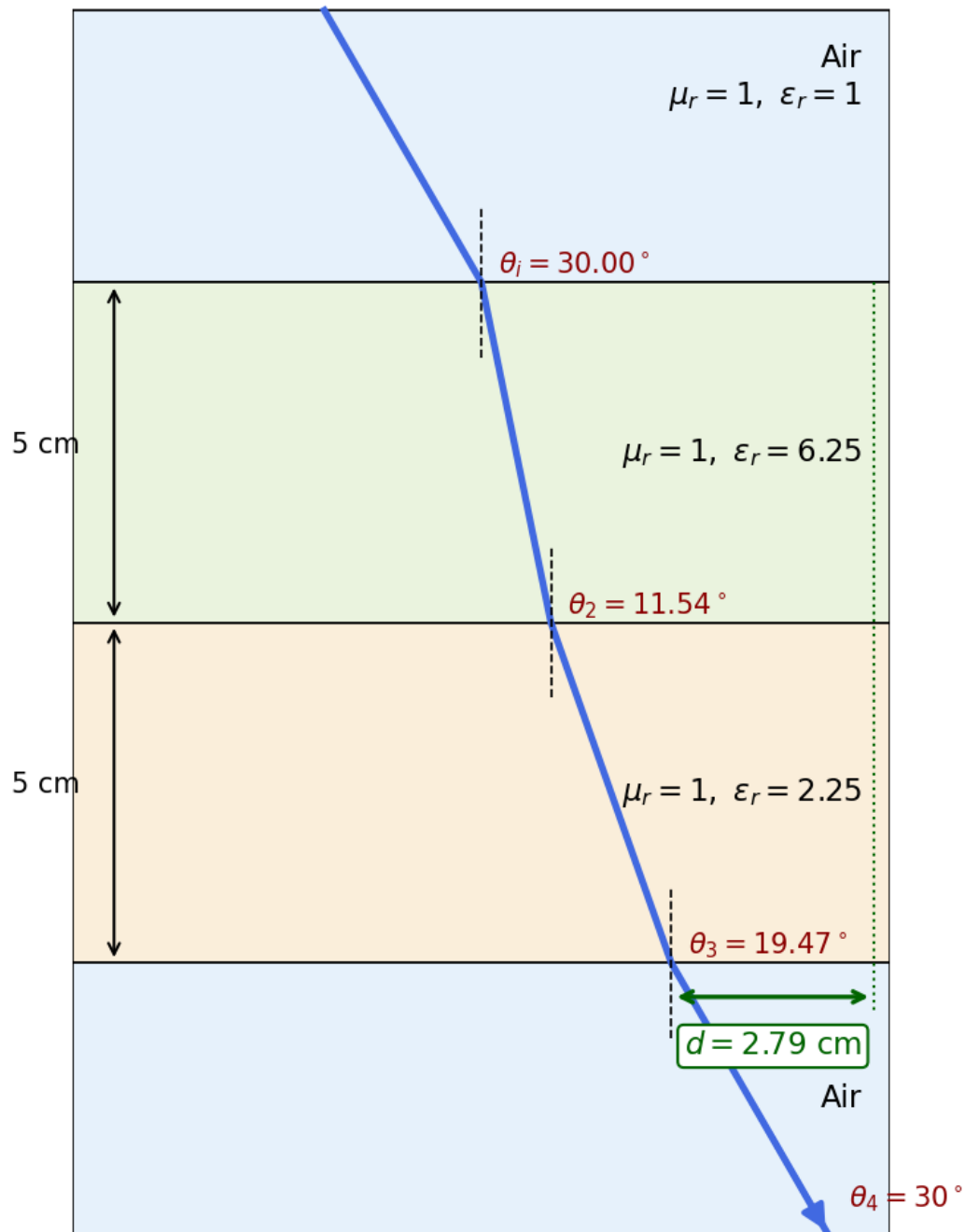
$$\sin \theta_4 = \frac{1}{3} \sqrt{\frac{2.25}{1}} = \frac{1}{3} \cdot \frac{3}{2} = \frac{1}{2} \Rightarrow \theta_4 = \boxed{30^\circ = \theta_i}$$

(b) Lateral displacement:

$$d = t_1 \tan \theta_2 + t_2 \tan \theta_3 = 5 \tan 11.54^\circ + 5 \tan 19.47^\circ = 5(0.2041 + 0.3536) \text{ cm}$$

$$\boxed{d = 2.79 \text{ cm}}$$

## HW3 P3: parallel-polarized wave through dielectric layers



## Problem 4 — Dielectric Waveguide TE Mode Identification

20 pts

**Setup:** TE wave,  $a = 5$  cm,  $b = 3$  cm,  $E_x = -36 \cos(40\pi x) \sin(100\pi y) \sin((2.4\pi \times 10^{10})t - 52.9\pi z)$  V/m.

## RECTANGULAR WAVEGUIDE — MODES

**TE mode**  $E_z = 0, H_z \neq 0$

**TM mode**  $H_z = 0, E_z \neq 0$

Dimensions  $a \times b$  with  $a > b$  along  $\hat{x}, \hat{y}$ ; wave in  $\hat{z}$ .

$E_x$  form (TE) read  $m, n$  from arguments

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$$

coef. on  $x = m\pi/a$ ; coef. on  $y = n\pi/b$ .

TE mode allowed if at least one of  $m, n \geq 1$ . TM needs both  $m, n \geq 1$ .

 $\epsilon_r$  FROM DISPERSION

Given  $\omega, \beta$ , mode  $(m, n)$

$$\epsilon_r = \frac{c^2}{\omega^2} \left[ \beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$$

Sanity:  $\beta$  in rad/m,  $a, b$  in m,  $\omega$  in rad/s.

## CUTOFF FREQUENCY

$f_{mn}$   $m, n$  integers  $\geq 0$  (not both zero)

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}$$

$$u_{p0} = \frac{c}{\sqrt{\epsilon_r}} \text{ (m/s, guide medium)}$$

Common cutoffs ( $a > b$ )

| Mode                                | $f_c$                                   |
|-------------------------------------|-----------------------------------------|
| TE <sub>10</sub>                    | $u_{p0}/(2a)$ (dominant)                |
| TE <sub>01</sub>                    | $u_{p0}/(2b)$                           |
| TE <sub>20</sub>                    | $u_{p0}/a = 2f_{10}$                    |
| TE <sub>11</sub> , TM <sub>11</sub> | $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$ |

Propagation requires  $f > f_{mn}$ .

WAVE IMPEDANCE &  $H_y$ 

$$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} \text{ (}\Omega\text{)} \quad Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$$

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \quad H_y = \frac{E_x}{Z_{TE}} \text{ (TE mode)}$$

$Z_{TE} > \eta$  always (denominator  $< 1$ );  $Z_{TM} < \eta$ .

(a) **Mode from the field expression.** Read  $m\pi/a$  from the cos coefficient and  $n\pi/b$  from the sin coefficient.

$$\frac{m\pi}{a} = 40\pi \Rightarrow m = 40 \cdot 0.05 = 2$$

$$\frac{n\pi}{b} = 100\pi \Rightarrow n = 100 \cdot 0.03 = 3$$

Mode: TE<sub>23</sub>

(b)  $\epsilon_r$  from dispersion, with  $\omega = 2.4\pi \times 10^{10}$  rad/s,  $\beta = 52.9\pi$  rad/m:

$$\epsilon_r = \frac{c^2}{\omega^2} \left[ \beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right] = \left( \frac{3 \times 10^8}{2.4\pi \times 10^{10}} \right)^2 [(52.9\pi)^2 + (40\pi)^2 + (100\pi)^2]$$

$$\epsilon_r = 2.25$$

(c) Cutoff frequency for TE<sub>23</sub>:

$$u_{p0} = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{1.5} = 2 \times 10^8 \text{ m/s}$$

$$f_{23} = \frac{u_{p0}}{2} \sqrt{\left(\frac{2}{a}\right)^2 + \left(\frac{3}{b}\right)^2} = 10^8 \sqrt{(2/0.05)^2 + (3/0.03)^2} = 10^8 \sqrt{1600 + 10000}$$

$$f_{23} = 10.77 \text{ GHz}$$

(d)  $H_y$  from  $Z_{TE}$ : operating frequency  $f = \omega/(2\pi) = 12$  GHz.

$$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} = \frac{377/\sqrt{2.25}}{\sqrt{1 - (10.77/12)^2}} = \frac{251.3}{\sqrt{1 - 0.805}} = \frac{251.3}{0.442} = 569.9 \text{ }\Omega$$

$$H_y = \frac{E_x}{Z_{TE}} = -0.063 \cos(40\pi x) \sin(100\pi y) \sin((2.4\pi \times 10^{10})t - 52.9\pi z) \text{ A/m}$$

**Problem 5 — Hollow Guide, Multi-Mode Excitation & Travel Times** 8 pts

**Setup:** Hollow guide ( $\epsilon_r = 1$ ,  $u_{p0} = c$ ),  $a = 3$  cm,  $b = 2$  cm. Pulse with 9.5 GHz carrier, guide length 100 m.

**CUTOFF FREQUENCY**

$f_{mn}$   $m, n$  integers  $\geq 0$  (not both zero)

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}$$

$$u_{p0} = \frac{c}{\sqrt{\epsilon_r}} \text{ (m/s, guide medium)}$$

Common cutoffs ( $a > b$ )

| Mode                                | $f_c$                                   |
|-------------------------------------|-----------------------------------------|
| TE <sub>10</sub>                    | $u_{p0}/(2a)$ (dominant)                |
| TE <sub>01</sub>                    | $u_{p0}/(2b)$                           |
| TE <sub>20</sub>                    | $u_{p0}/a = 2f_{10}$                    |
| TE <sub>11</sub> , TM <sub>11</sub> | $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$ |

Propagation requires  $f > f_{mn}$ .

**GROUP VELOCITY & TRAVEL TIME**

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s)}$$

$$u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}} \Rightarrow u_p u_g = u_{p0}^2$$

$t = L/u_g$  (s) travel time through guide of length  $L$   
Higher modes have lower  $u_g \Rightarrow$  they arrive later. Group delay spread = sum over excited modes.

**Cutoff frequencies for the lowest modes:**

$$f_{10} = \frac{c}{2a} = \frac{3 \times 10^8}{0.06} = 5 \text{ GHz}$$

$$f_{01} = \frac{c}{2b} = \frac{3 \times 10^8}{0.04} = 7.5 \text{ GHz}$$

$$f_{11} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}} = (1.5 \times 10^8) \sqrt{(1/0.03)^2 + (1/0.02)^2} = 9.01 \text{ GHz}$$

$$f_{20} = \frac{c}{a} = 10 \text{ GHz} \quad (> 9.5, \text{ not excited})$$

**Modes excited (carrier  $\hat{f}_c$ ):** TE<sub>10</sub>, TE<sub>01</sub>, TE<sub>11</sub>, TM<sub>11</sub>. TE<sub>20</sub> is just above cutoff so it does not propagate.

**Group velocities:**  $u_g = c \sqrt{1 - (f_{mn}/f)^2}$  with  $f = 9.5$  GHz.

| Mode                                | $f_c$ (GHz) | $u_g/c$                          | $u_g$ (m/s)        |
|-------------------------------------|-------------|----------------------------------|--------------------|
| TE <sub>10</sub>                    | 5.00        | $\sqrt{1 - (5/9.5)^2} = 0.85$    | $2.55 \times 10^8$ |
| TE <sub>01</sub>                    | 7.50        | $\sqrt{1 - (7.5/9.5)^2} = 0.61$  | $1.84 \times 10^8$ |
| TE <sub>11</sub> , TM <sub>11</sub> | 9.01        | $\sqrt{1 - (9.01/9.5)^2} = 0.32$ | $0.95 \times 10^8$ |

**Travel times:**  $t = L/u_g$  over 100 m.

| Mode                                | $t$ ( $\mu$ s) |
|-------------------------------------|----------------|
| TE <sub>10</sub>                    | 0.39           |
| TE <sub>01</sub>                    | 0.54           |
| TE <sub>11</sub> , TM <sub>11</sub> | 1.05           |

Higher modes arrive later because they propagate closer to cutoff, so  $u_g$  drops.